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UNIT II

Chapter 4

Organ and Organ Systems in Animals

Chapter Outline

4.1 Earthworm
4.2 Cockroach
4.3 Frog
4.4 Pigeon



A function to each organ and each organ to its own function is seen in all animals.

Learning Objectives:

- *Understands and appreciates the morphology of the earthworm, cockroach and frog.*
- *Recognises the functions of different organ systems.*
- *Appreciates the differences in the structural organization of the earthworm, cockroach frog and Pigeon*

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Introduction

From microbes to the blue whale, organisms occur in different sizes and shapes with a well organized organ and organ systems. The basic tissues (chapter-3) organize to form an organ which in turn associates to form organ systems in multicellular organisms. Such an organization is essential for efficient and better coordinated activities of millions of cells constituting an organism. You are being introduced to understand the morphology and anatomy of three organisms placed at different evolutionary levels to show their organization and functions. Morphology refers to the study of form or externally visible features. The word anatomy is used for the study of internal organs in the animals. This chapter deals with the morphology and anatomy of invertebrates represented by the earthworm and cockroach and the vertebrates represented by the frog and pigeon.

4.1 Earthworm - *Lampito mauritii*

Earthworm is a terrestrial invertebrate that inhabits the upper layers of the moist soil, rich in decaying organic matter. It is nocturnal and during the day it lives in burrows made by burrowing and swallowing the soil. In gardens, they can be traced by their faecal deposits known as worm castings on the soil surface. Earthworms are considered as **"Friends of Farmers".** The common Indian earthworms are *Lampito mauritii* (Syn. *Megascolex mauritii*), *Perioynx excavatus* and *Metaphire posthuma* (Syn.